HUMAnN3 Functional Profiling with Differential Analysis

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Overview

This practical introduces students to HUMAnN3-derived functional microbiome outputs using pathway abundance, pathway coverage, and gene family tables.

Questions addressed in this session

1. What microbial functions are most abundant across the cohort?
2. Do samples cluster by clinical group based on functional profiles?
3. Which broad biological categories dominate the microbiome?
4. Which pathways differ across groups?
5. Which gene families differ across groups?
6. Do the results highlight biology linked to the paper, such as metabolism, respiration, iron/heme handling, and nutrient utilization?

Expected learning points

- HUMAnN outputs include both broad functional summaries and high-resolution gene family features.
- Pathway abundance and pathway coverage answer different questions.
- Gene-family analysis requires filtering because of the very large number of features.
- Exploratory plots show broad structure, while differential abundance analysis identifies specific group-associated functions.

Prerequisites

This module assumes:

- Basic familiarity with R and RStudio
- Completed upstream processing (e.g., BWA, Prokka, Sylph, MetaPhlAn, HUMAnN3)
- Availability of:
  - pathway abundance tables
  - pathway coverage tables
  - gene family tables
  - sample metadata

Required R packages: - tidyverse - pheatmap - vegan - Maaslin2

Input data structure

- Rows = features (pathways or gene families)
- Columns = samples
- First column = Feature IDs
- Metadata must include:
  - sample_name
  - run_accession
  - group

```r
# ============================================
# LIBRARIES
# ============================================

library(tidyverse)
library(pheatmap)
library(vegan)

if (!requireNamespace("Maaslin2", quietly = TRUE)) {
  if (!requireNamespace("BiocManager", quietly = TRUE)) {
    install.packages("BiocManager")
  }
  BiocManager::install("Maaslin2")
}
library(Maaslin2)
```

```r
# ============================================
# HELPER-FUNCTIONS
# ============================================

extract_run_id <- function(x) {
  x <- gsub("_Abundance-RPKs$", "", x)
  x <- gsub("_Abundance$", "", x)
  x <- gsub("_Coverage$", "", x)
  x
}

clean_names <- function(x) {
  x <- gsub("^.*?:\\s*", "", x)   # remove pathway ID prefix
  x <- gsub("\\(.*?\\)", "", x)   # remove bracketed text like (E. coli)
  x <- gsub("_", " ", x)
  trimws(x)
}

annotate_pathways <- function(x) {
  case_when(
    grepl("glycolysis|fermentation|respiration|TCA|energy|formate", x, ignore.case = TRUE) ~ "Energy met
    grepl("amino acid|valine|leucine|isoleucine|ornithine|biosynthesis", x, ignore.case = TRUE) ~ "Amino
    grepl("nucleotide|purine|pyrimidine|UMP|adenine|guanosine|allantoin", x, ignore.case = TRUE) ~ "Nucl
    grepl("peptidoglycan|cell wall|muramoyl|lipopolysaccharide", x, ignore.case = TRUE) ~ "Cell wall",
    grepl("carbohydrate|glycogen|sugar|fucose|galactarate", x, ignore.case = TRUE) ~ "Carbohydrate metal
    grepl("cofactor|vitamin|heme|iron|aerobactin|ubiquinol|menaquinol", x, ignore.case = TRUE) ~ "Cofact
    TRUE ~ "Other"
```

```r
  )
}

remove_zero_variance_features <- function(mat) {
  keep <- apply(mat, 1, function(x) {
    x <- x[!is.na(x)]
    length(x) > 1 && var(x) > 0
  })
  mat[keep, , drop = FALSE]
}
```

```r
# ============================================
# LOAD DATA
# ============================================


path_abun <- read.delim(
  "/Users/LuicerAnneIngasia/Desktop/wcs/humman/humann3_pathabundance_merged_normalised_UG.tsv",
  sep = "\t",
  check.names = FALSE
)

path_cov <- read.delim(
  "/Users/LuicerAnneIngasia/Desktop/wcs/humman/humann3_pathcoverage_merged_UG.tsv",
  sep = "\t",
  check.names = FALSE
)

genefam <- read.delim(
  "/Users/LuicerAnneIngasia/Desktop/wcs/humman/humann3_genefamilies_merged_normalised_UG-2.tsv",
  sep = "\t",
  check.names = FALSE
)

meta <- read.csv(
  "/Users/LuicerAnneIngasia/Desktop/wcs/cleaned_metadata.csv",
  check.names = FALSE
)

dim(path_abun)
```

```
## [1] 22065    38
```

```r
dim(path_cov)
```

```
## [1] 19873    25
```

```r
dim(genefam)
```

```
## [1] 3278640      38
```

```r
dim(meta)
```

```
## [1] 41 58
```

```r
head(meta)
```

```
##         sample sample_name run_accession country studyid uric_se1 group
## 1 CM_1_met_abs    CM_1_S71   SRR30598656  Uganda    1109      1.8     1
## 2 CM_2_met_abs    CM_2_S72          <NA>  Malawi    1199      3.7     1
## 3 CM_3_met_abs    CM_3_S73   SRR30598651  Uganda    1384      6.8     1
## 4 CM_4_met_abs    CM_4_S74   SRR30598648  Uganda    1389      8.5     1
## 5 CM_5_met_abs    CM_5_S75   SRR30598640  Uganda    1450      3.0     1
## 6 CM_6_met_abs    CM_6_S76   SRR30598628  Uganda    1600      5.7     1
##   acidotic ceftri ceftri1 ceftri2 antibiot2 antibiot1 antibio2 chloram cipro
## 1        0      0       1       0         0         1        0       0     0
## 2        0      0       1       0         0         1        0       0     0
## 3        0      0       1       0         0         1        0       0     0
## 4        0      0       1       0         1         1        0       0     0
## 5        0      0       1       0         1         1        0       0     0
## 6        0      0       1       0         0         1        0       0     0
##   xpen gentam flagyl cotrimoxazole        doe  stool_dt0 chloram1 cipro1 xpen1
## 1    0      0      0             0  3/24/2015  3/28/2015        0      0     0
## 2    0      0      0             0  3/22/2016  3/24/2016        0      0     0
## 3    0      0      0             0 03/03/2016 03/07/2016        0      0     0
## 4    0      0      0             0  2/24/2016  2/25/2016        0      0     0
## 5    0      0      0             0  9/17/2016  9/21/2016        0      0     0
## 6    0      0      0             0  1/26/2017  1/26/2017        0      0     0
##   gentam1 flagyl1 amoxyl lacidosis tff3_pl1 outcome sm_death12m site sex  age
## 1       0       0      0         0    1.259       0           0    1   1 3.46
## 2       1       0      0         0    1.774       0           0    2   1 3.97
## 3       0       0      0         0    2.184       0           0    1   1 3.67
## 4       0       0      0         0    1.849       0           0    1   2 3.86
## 5       0       0      0         0    1.674       0           0    1   1 3.87
## 6       0       0      0         0    2.143       0           0    1   2 3.09
##   hbsgt haki inpatmal12m inpatmalno12m inpatany12m enr_sma hgbbase feedhour
## 1     1    0           0             0           0       0     8.5       48
## 2     1    0           1             1           1       0     9.6       53
## 3     1    0           1             1           1       0     7.4       32
## 4     1    0           0             0           0       0     7.6       24
## 5     1    0           0             0           0       0     7.3       18
## 6     1    0          NA            NA          NA       0     7.3       24
##   neuabs  ho1_pl1 blcx_pos   wbc        dod trem_pl1 inpatsepsis12m
## 1   4.26  61.5300        0  6.93  3/30/2015   278.35              0
## 2   4.17  58.8200        0  5.84 04/05/2016   425.56              0
## 3   5.57 113.8275        0  6.79 03/09/2016   356.35              0
## 4   9.61  73.6125        0 15.87  2/28/2016   265.79              0
## 5   8.46 102.0850        0 13.22  9/22/2016   197.67              0
## 6   3.24   7.9700        0  6.26  1/31/2017   203.40             NA
##   deathdaysto12m blcx_result blcx_result_spec othantibi  whz0  haz0  waz0
## 1             NA          NA               NA        NA -0.78 -2.54 -1.96
## 2             NA          NA               NA        NA -2.03 -0.29 -1.49
## 3             NA          NA               NA        NA  0.71 -2.58 -1.03
## 4             NA           6               NA        NA -1.18 -1.09 -1.45
```

```
## 5             NA          NA               NA        NA -0.99 -2.91 -2.35
## 6             NA          NA               NA        NA -1.38 -1.16 -1.59
##   pcr_fal0
## 1        1
## 2        1
## 3        1
## 4        1
## 5        1
## 6        1
```

```r
# ============================================
# CLEAN DATA
# ============================================

colnames(path_abun)[1] <- "Feature"
colnames(path_cov)[1]  <- "Feature"
colnames(genefam)[1]   <- "Feature"

path_abun <- path_abun %>%
  filter(!grepl("UNMAPPED|UNINTEGRATED", Feature))

path_cov <- path_cov %>%
  filter(!grepl("UNMAPPED|UNINTEGRATED", Feature))

genefam <- genefam %>%
  filter(!grepl("UNMAPPED|UNINTEGRATED", Feature))

# Keep unstratified rows for main functional analysis
path_abun_unstrat <- path_abun %>%
  filter(!grepl("\\|", Feature))

path_cov_unstrat <- path_cov %>%
  filter(!grepl("\\|", Feature))
```

```r
# ============================================
# CHECK COLUMN NAMING
# ============================================

head(colnames(path_abun_unstrat))
```

```
## [1] "Feature"               "SRR30598618_Abundance" "SRR30598619_Abundance"
## [4] "SRR30598620_Abundance" "SRR30598621_Abundance" "SRR30598622_Abundance"
```

```r
head(colnames(path_cov_unstrat))
```

```
## [1] "Feature"              "SRR30598619_Coverage" "SRR30598621_Coverage"
## [4] "SRR30598622_Coverage" "SRR30598625_Coverage" "SRR30598626_Coverage"
```

```r
head(colnames(genefam))
```

```
## [1] "Feature"                     "SRR30598618_Abundance-RPKs"
## [3] "SRR30598619_Abundance-RPKs" "SRR30598620_Abundance-RPKs"
## [5] "SRR30598621_Abundance-RPKs" "SRR30598622_Abundance-RPKs"
```

```r
ncol(path_abun_unstrat)
```

```
## [1] 38
```

```r
ncol(path_cov_unstrat)
```

```
## [1] 25
```

```r
ncol(genefam)
```

```
## [1] 38
```

```r
# ============================================
# MATCH METADATA TO HUMAnN SAMPLES
# ============================================

meta2 <- meta %>%
  filter(!is.na(run_accession), run_accession != "")

abun_ids <- extract_run_id(colnames(path_abun_unstrat)[-1])
cov_ids  <- extract_run_id(colnames(path_cov_unstrat)[-1])
gene_ids <- extract_run_id(colnames(genefam)[-1])

cat("Pathway abundance samples:", length(abun_ids), "\n")
```

```
## Pathway abundance samples: 37
```

```r
cat("Pathway coverage samples:", length(cov_ids), "\n")
```

```
## Pathway coverage samples: 24
```

```r
cat("Gene family samples:", length(gene_ids), "\n")
```

```
## Gene family samples: 37
```

```r
cat("Metadata samples with run_accession:", nrow(meta2), "\n")
```

```
## Metadata samples with run_accession: 24
```

```r
shared_ids <- Reduce(intersect, list(abun_ids, cov_ids, gene_ids, meta2$run_accession))
shared_ids <- abun_ids[abun_ids %in% shared_ids]

cat("Shared samples across all data types and metadata:", length(shared_ids), "\n")
```

```
## Shared samples across all data types and metadata: 24
```

```r
abun_keep <- match(shared_ids, abun_ids)
cov_keep  <- match(shared_ids, cov_ids)
gene_keep <- match(shared_ids, gene_ids)

path_abun_unstrat <- path_abun_unstrat[, c(1, abun_keep + 1), drop = FALSE]
path_cov_unstrat  <- path_cov_unstrat[, c(1, cov_keep + 1), drop = FALSE]
genefam           <- genefam[, c(1, gene_keep + 1), drop = FALSE]

meta2 <- meta2[match(shared_ids, meta2$run_accession), , drop = FALSE]

stopifnot(all(extract_run_id(colnames(path_abun_unstrat)[-1]) == shared_ids))
stopifnot(all(extract_run_id(colnames(path_cov_unstrat)[-1]) == shared_ids))
stopifnot(all(extract_run_id(colnames(genefam)[-1]) == shared_ids))
stopifnot(all(meta2$run_accession == shared_ids))

meta <- meta2
meta$group <- as.factor(meta$group)

colnames(path_abun_unstrat)[-1] <- meta$sample_name
colnames(path_cov_unstrat)[-1]  <- meta$sample_name
colnames(genefam)[-1]           <- meta$sample_name

cat("Final matched sample count:", nrow(meta), "\n")
```

```
## Final matched sample count: 24
```

```r
table(meta$group)
```

```
##
## 1 2 4 6
## 7 7 5 5
```

```r
head(meta[, c("sample_name", "run_accession", "group", "country")])
```

```
##    sample_name run_accession group country
## 23   SMA_6_S85   SRR30598619     4  Uganda
## 19  RDS_8_S100   SRR30598621     2  Uganda
## 20   SMA_2_S81   SRR30598622     4  Uganda
## 24   SMA_7_S86   SRR30598625     4  Uganda
## 14   RDS_2_S94   SRR30598626     2  Uganda
## 13   RDS_1_S93   SRR30598627     2  Uganda
```

Question 1: What are the most abundant microbial functions across the cohort?

These first plots are exploratory. They help us understand broad functional structure before testing specific associations.

```r
# ============================================
# PATHWAY MATRIX
# ============================================
```

```r
path_mat <- path_abun_unstrat %>%
  column_to_rownames("Feature") %>%
  as.matrix()

mode(path_mat) <- "numeric"
path_mat_log <- log10(path_mat + 1e-6)
```

We begin by examining global structure using dimensionality reduction. Principal Component Analysis (PCA) allows us to assess whether samples cluster by group.

```r
# ============================================
# PCA OF PATHWAY ABUNDANCE
# ============================================

pca_input <- t(path_mat_log)

keep_cols <- apply(
  pca_input,
  2,
  function(x) {
    x <- x[!is.na(x)]
    length(x) > 1 && var(x) > 0
  }
)

pca_input <- pca_input[, keep_cols, drop = FALSE]

cat("Samples used for PCA:", nrow(pca_input), "\n")
```

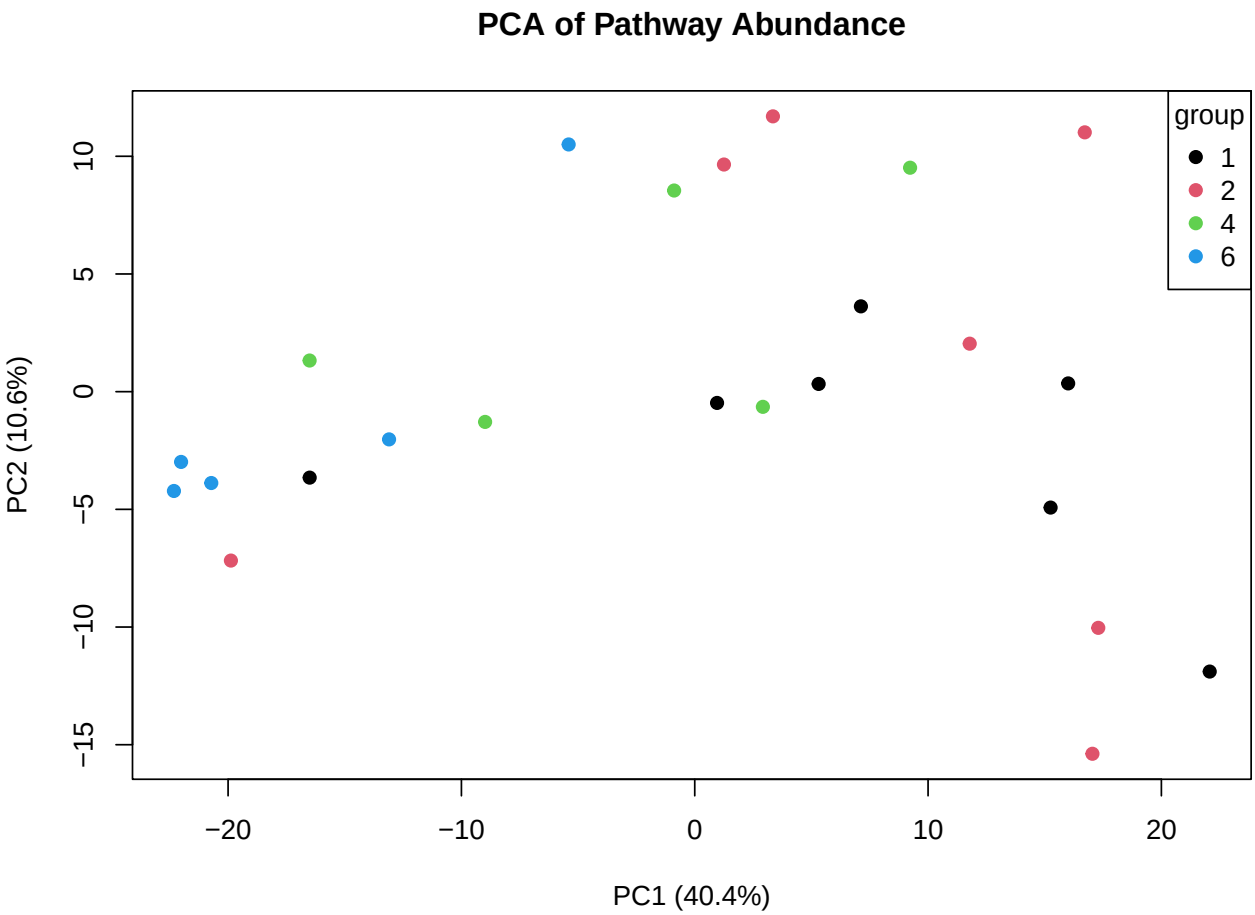
```
## Samples used for PCA: 24
```

```r
cat("Pathways retained for PCA:", ncol(pca_input), "\n")
```

```
## Pathways retained for PCA: 517
```

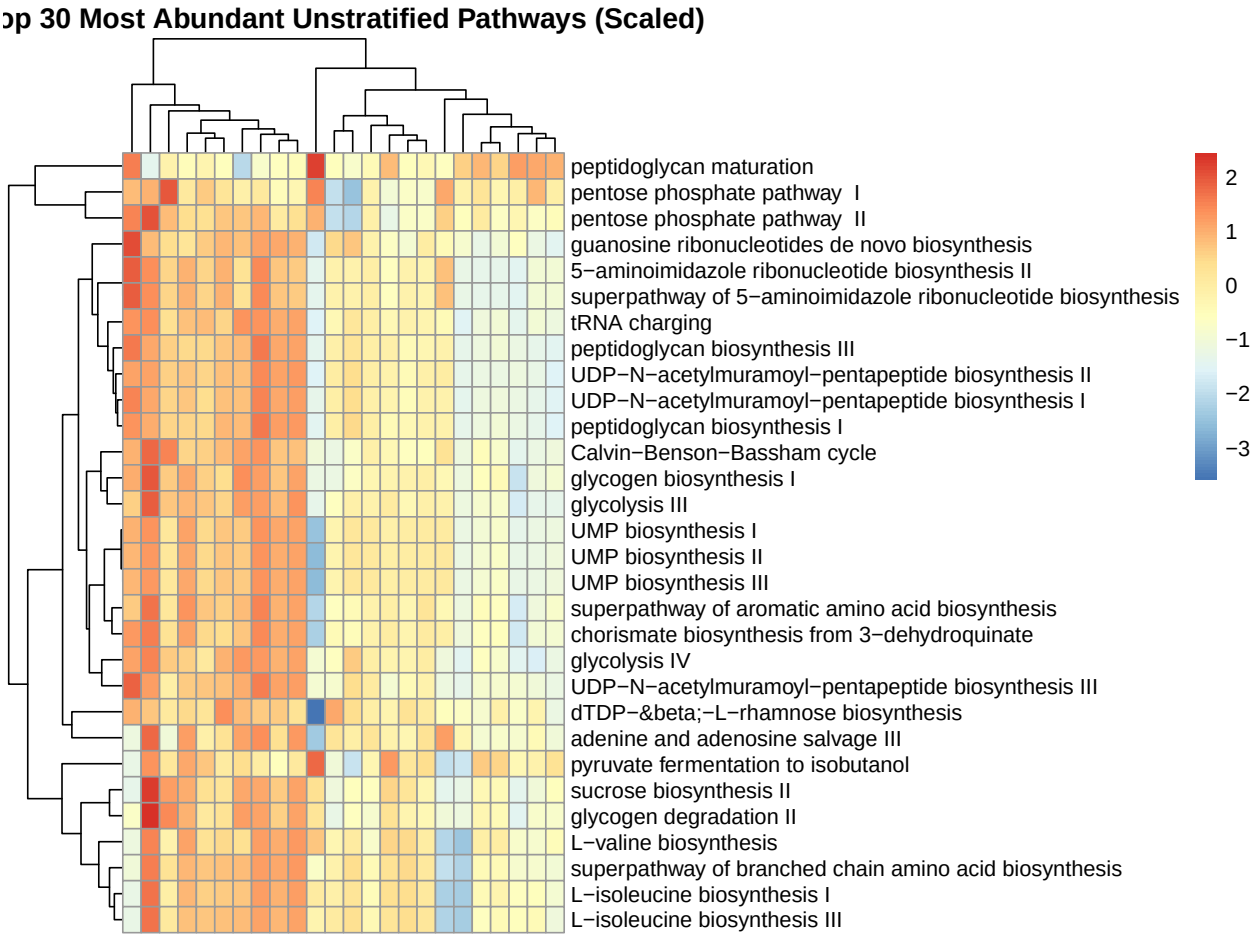
```r
pca <- prcomp(pca_input, scale. = TRUE)

plot(
  pca$x[, 1], pca$x[, 2],
  col = meta$group,
  pch = 19,
  xlab = paste0("PC1 (", round(100 * summary(pca)$importance[2, 1], 1), "%)"),
  ylab = paste0("PC2 (", round(100 * summary(pca)$importance[2, 2], 1), "%)"),
  main = "PCA of Pathway Abundance"
)

legend(
  "topright",
  legend = levels(meta$group),
  col = seq_along(levels(meta$group)),
  pch = 19,
```

```
  title = "group"
)
```



Interpretation prompt:
Do samples separate clearly by group, or is the clustering partial? If separation is incomplete, that suggests the functional shifts are subtle rather than absolute.

```
# ==========================================
# TOP ABUNDANT PATHWAYS
# ==========================================

path_means <- rowMeans(path_mat, na.rm = TRUE)
top_pathways <- names(sort(path_means, decreasing = TRUE))[1:30]

heat_path <- path_mat_log[top_pathways, , drop = FALSE]
heat_path_scaled <- t(scale(t(heat_path)))

rownames(heat_path_scaled) <- clean_names(rownames(heat_path_scaled))

pheatmap(
  heat_path_scaled,
  show_rownames = TRUE,
  show_colnames = FALSE,
  main = "Top 30 Most Abundant Unstratified Pathways (Scaled)"
)
```



Interpretation prompt:
These pathways usually reflect core metabolism and microbial growth rather than group-specific biology.

Question 2: How consistently are pathways detected across samples?

```
# ============================================
# PATHWAY COVERAGE
# ============================================

cov_mat <- path_cov_unstrat %>%
  column_to_rownames("Feature") %>%
  as.matrix()

mode(cov_mat) <- "numeric"

prev <- rowSums(cov_mat > 0, na.rm = TRUE)
cov_filtered <- cov_mat[prev > 0.2 * ncol(cov_mat), , drop = FALSE]

coverage_summary <- data.frame(
  Feature = rownames(cov_mat),
  Prevalence = prev,
  Proportion = prev / ncol(cov_mat)
```

```r
) %>%
  arrange(desc(Proportion))

dim(cov_filtered)
```

```
## [1] 128  24
```

```r
head(coverage_summary, 20)
```

```
##
## ASPASN-PWY: superpathway of L-aspartate and L-asparagine biosynthesis
## CALVIN-PWY: Calvin-Benson-Bassham cycle
## GLUCOSE1PMETAB-PWY: glucose and glucose-1-phosphate degradation
## GLYCOGENSYNTH-PWY: glycogen biosynthesis I (from ADP-D-Glucose)
## ILEUSYN-PWY: L-isoleucine biosynthesis I (from threonine)
## NONOXIPENT-PWY: pentose phosphate pathway (non-oxidative branch) I
## PEPTIDOGLYCANSYN-PWY: peptidoglycan biosynthesis I (meso-diaminopimelate containing)
## PWY-5103: L-isoleucine biosynthesis III
## PWY-5667: CDP-diacylglycerol biosynthesis I
## PWY-5695: inosine 5'-phosphate degradation
## PWY-5941: glycogen degradation II
## PWY-6121: 5-aminoimidazole ribonucleotide biosynthesis I
## PWY-6122: 5-aminoimidazole ribonucleotide biosynthesis II
## PWY-6277: superpathway of 5-aminoimidazole ribonucleotide biosynthesis
## PWY-6385: peptidoglycan biosynthesis III (mycobacteria)
## PWY-6386: UDP-N-acetylmuramoyl-pentapeptide biosynthesis II (lysine-containing)
## PWY-6387: UDP-N-acetylmuramoyl-pentapeptide biosynthesis I (meso-diaminopimelate containing) PWY-638
## PWY-6609: adenine and adenosine salvage III
## PWY-7111: pyruvate fermentation to isobutanol (engineered)
## PWY-7221: guanosine ribonucleotides de novo biosynthesis
##                                                                                                Prevale
## ASPASN-PWY: superpathway of L-aspartate and L-asparagine biosynthesis
## CALVIN-PWY: Calvin-Benson-Bassham cycle
## GLUCOSE1PMETAB-PWY: glucose and glucose-1-phosphate degradation
## GLYCOGENSYNTH-PWY: glycogen biosynthesis I (from ADP-D-Glucose)
## ILEUSYN-PWY: L-isoleucine biosynthesis I (from threonine)
## NONOXIPENT-PWY: pentose phosphate pathway (non-oxidative branch) I
## PEPTIDOGLYCANSYN-PWY: peptidoglycan biosynthesis I (meso-diaminopimelate containing)
## PWY-5103: L-isoleucine biosynthesis III
## PWY-5667: CDP-diacylglycerol biosynthesis I
## PWY-5695: inosine 5'-phosphate degradation
## PWY-5941: glycogen degradation II
## PWY-6121: 5-aminoimidazole ribonucleotide biosynthesis I
## PWY-6122: 5-aminoimidazole ribonucleotide biosynthesis II
## PWY-6277: superpathway of 5-aminoimidazole ribonucleotide biosynthesis
## PWY-6385: peptidoglycan biosynthesis III (mycobacteria)
## PWY-6386: UDP-N-acetylmuramoyl-pentapeptide biosynthesis II (lysine-containing)
## PWY-6387: UDP-N-acetylmuramoyl-pentapeptide biosynthesis I (meso-diaminopimelate containing)
## PWY-6609: adenine and adenosine salvage III
## PWY-7111: pyruvate fermentation to isobutanol (engineered)
## PWY-7221: guanosine ribonucleotides de novo biosynthesis
##                                                                                                Proporti
```

```
## ASPASN-PWY: superpathway of L-aspartate and L-asparagine biosynthesis
## CALVIN-PWY: Calvin-Benson-Bassham cycle
## GLUCOSE1PMETAB-PWY: glucose and glucose-1-phosphate degradation
## GLYCOGENSYNTH-PWY: glycogen biosynthesis I (from ADP-D-Glucose)
## ILEUSYN-PWY: L-isoleucine biosynthesis I (from threonine)
## NONOXIPENT-PWY: pentose phosphate pathway (non-oxidative branch) I
## PEPTIDOGLYCANSYN-PWY: peptidoglycan biosynthesis I (meso-diaminopimelate containing)
## PWY-5103: L-isoleucine biosynthesis III
## PWY-5667: CDP-diacylglycerol biosynthesis I
## PWY-5695: inosine 5'-phosphate degradation
## PWY-5941: glycogen degradation II
## PWY-6121: 5-aminoimidazole ribonucleotide biosynthesis I
## PWY-6122: 5-aminoimidazole ribonucleotide biosynthesis II
## PWY-6277: superpathway of 5-aminoimidazole ribonucleotide biosynthesis
## PWY-6385: peptidoglycan biosynthesis III (mycobacteria)
## PWY-6386: UDP-N-acetylmuramoyl-pentapeptide biosynthesis II (lysine-containing)
## PWY-6387: UDP-N-acetylmuramoyl-pentapeptide biosynthesis I (meso-diaminopimelate containing)
## PWY-6609: adenine and adenosine salvage III
## PWY-7111: pyruvate fermentation to isobutanol (engineered)
## PWY-7221: guanosine ribonucleotides de novo biosynthesis
```

Interpretation prompt:
Pathway abundance asks how much of a pathway is present. Pathway coverage asks how consistently a pathway is represented across samples.

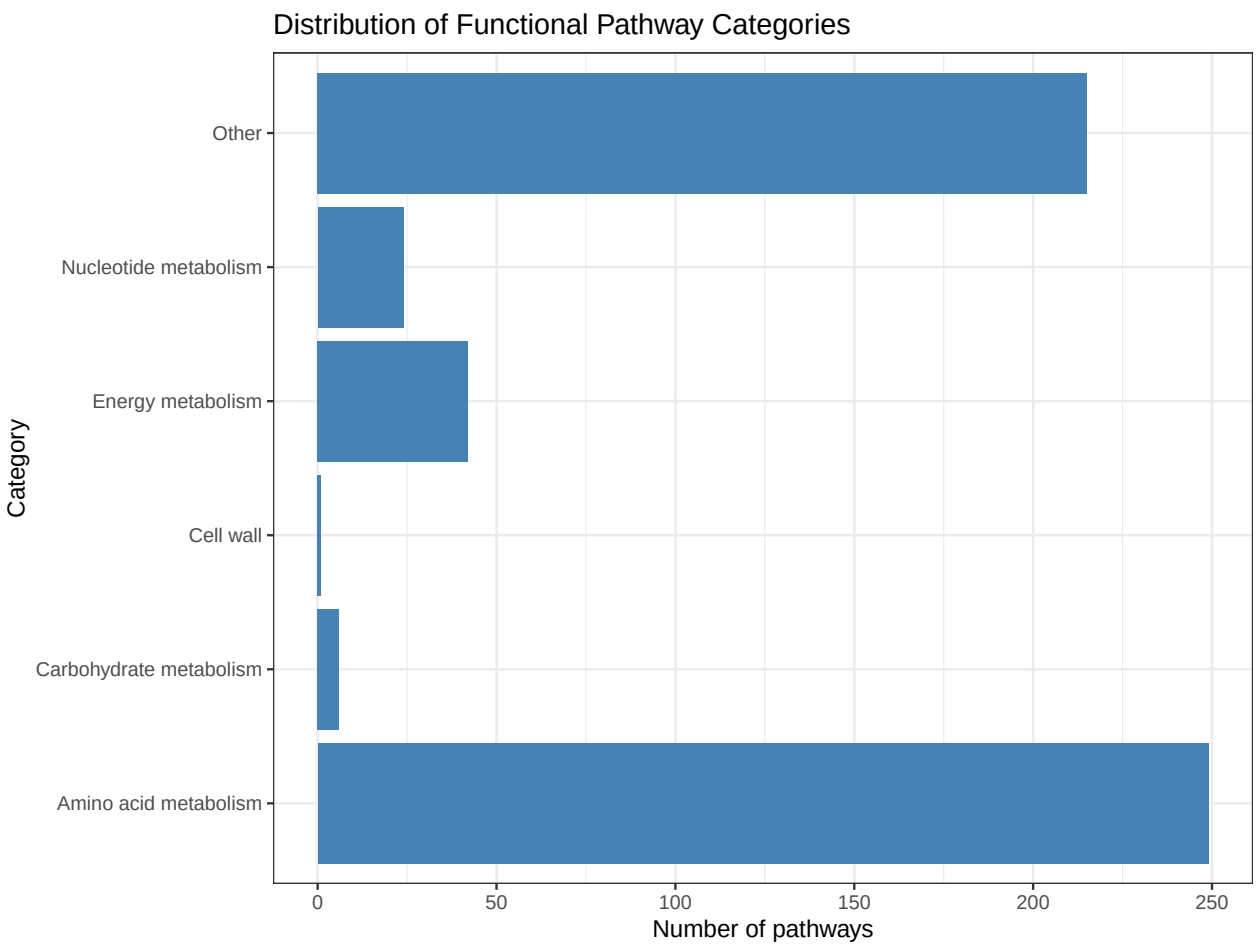
Question 3: Which broad biological systems dominate the microbiome?

```r
# ============================================
# FUNCTIONAL CATEGORIES
# ============================================

pathway_names <- clean_names(rownames(path_mat))
categories <- annotate_pathways(pathway_names)

table(categories)
```

```
## categories
##   Amino acid metabolism Carbohydrate metabolism               Cell wall
##                     249                       6                       1
##       Energy metabolism   Nucleotide metabolism                   Other
##                      42                      24                     215
```

```r
cat_df <- data.frame(category = categories)

ggplot(cat_df, aes(x = category)) +
  geom_bar(fill = "steelblue") +
  theme_bw() +
  coord_flip() +
  labs(
    title = "Distribution of Functional Pathway Categories",
    x = "Category",
```

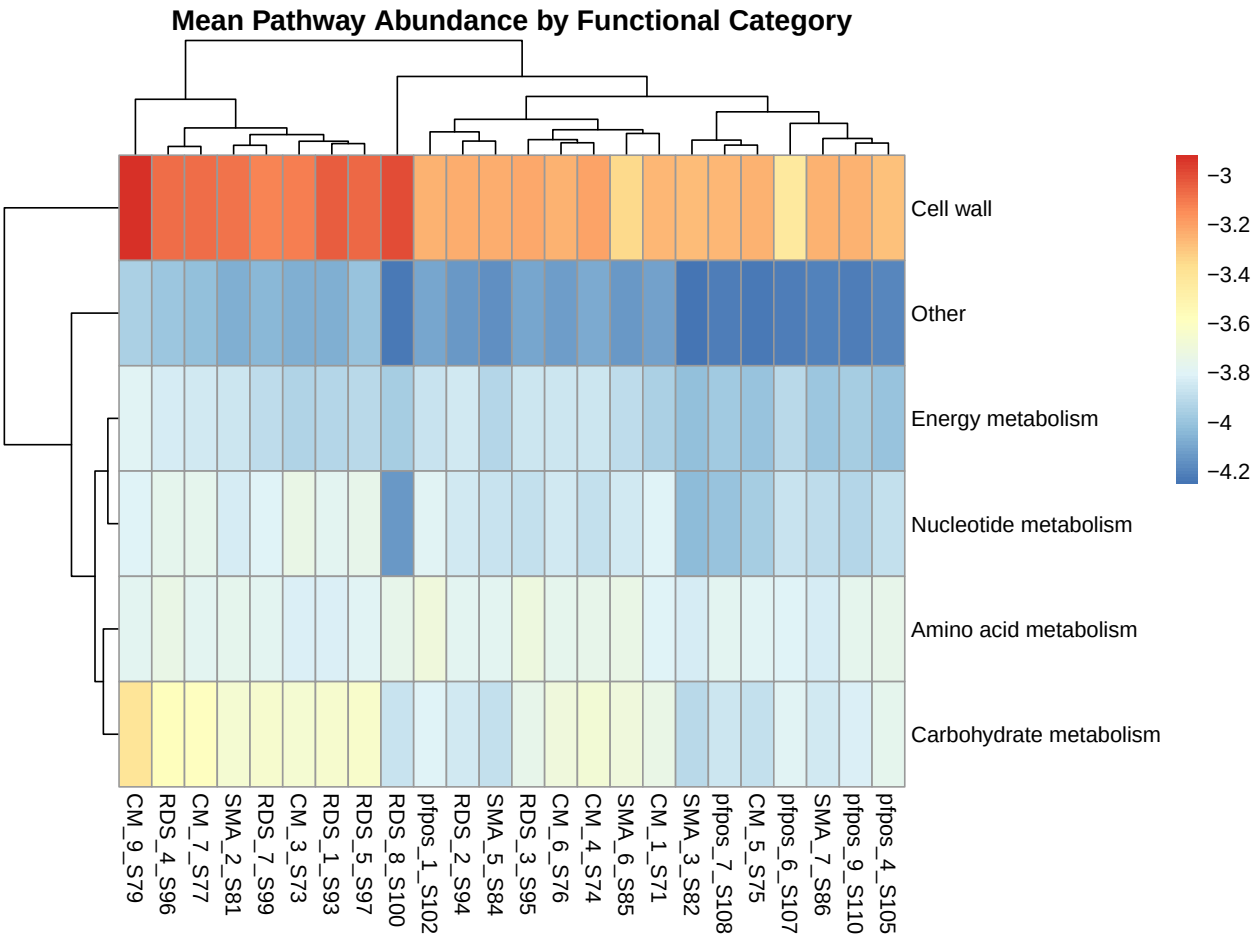
```
    y = "Number of pathways"
  )
```



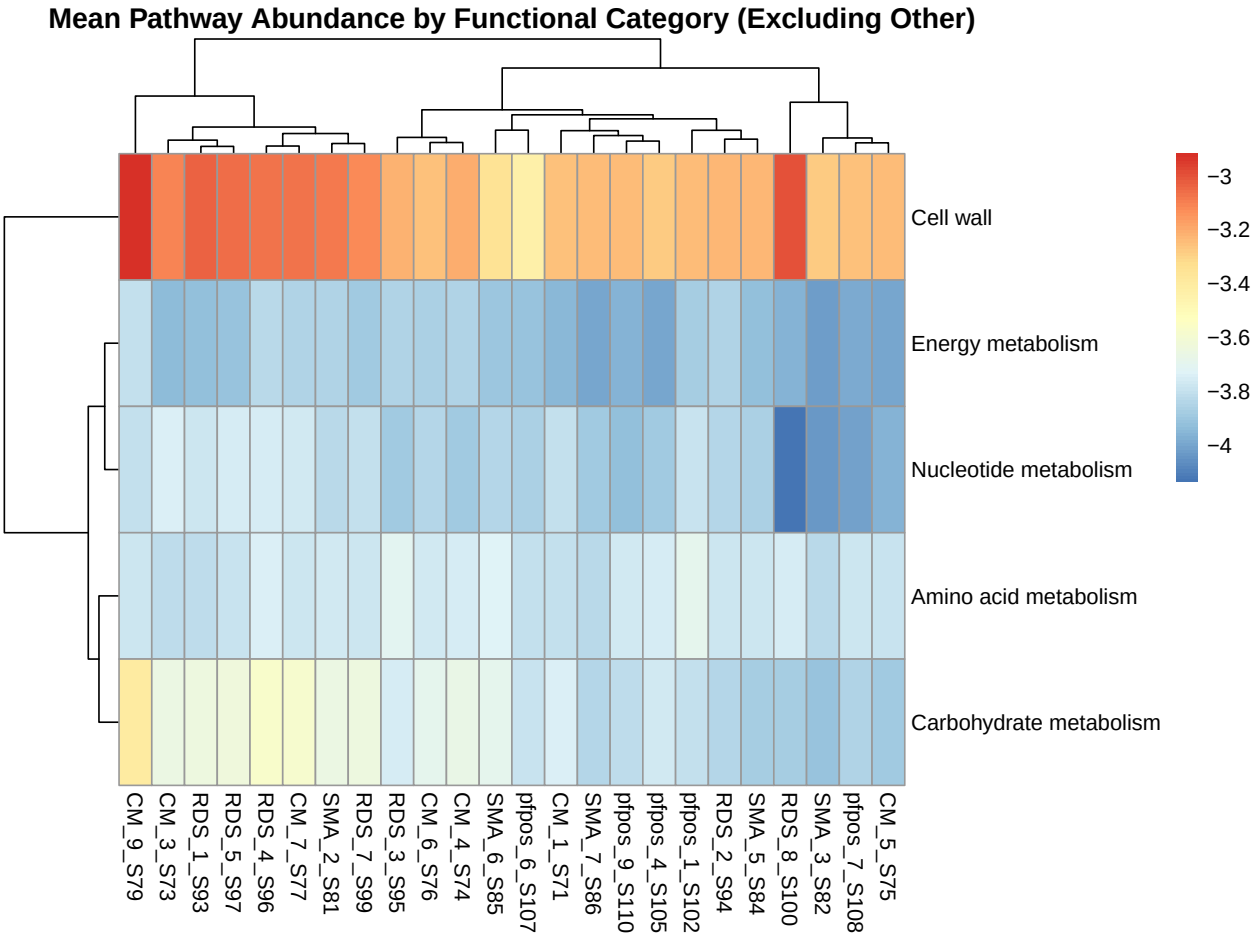
Interpretation prompt:
This plot shows how many pathways fall into each broad biological category. It helps move from pathway IDs to interpretable biology.

```
# ============================================
# MEAN ABUNDANCE BY FUNCTIONAL CATEGORY
# ============================================

path_df <- as.data.frame(path_mat)
path_df$category <- categories

category_abundance_mean <- path_df %>%
  group_by(category) %>%
  summarise(across(where(is.numeric), mean))

cat_mat_mean <- category_abundance_mean %>%
  column_to_rownames("category") %>%
  as.matrix()

cat_mat_mean_log <- log10(cat_mat_mean + 1e-6)
```

```r
pheatmap(
  cat_mat_mean_log,
  main = "Mean Pathway Abundance by Functional Category",
  cluster_rows = TRUE,
  cluster_cols = TRUE
)
```




```r
# ==========================================
# EXCLUDING THE BROAD 'OTHER' CATEGORY
# ==========================================

cat_mat_mean_no_other <- cat_mat_mean[rownames(cat_mat_mean) != "Other", , drop = FALSE]

pheatmap(
  log10(cat_mat_mean_no_other + 1e-6),
  main = "Mean Pathway Abundance by Functional Category (Excluding Other)",
  cluster_rows = TRUE,
  cluster_cols = TRUE
)
```

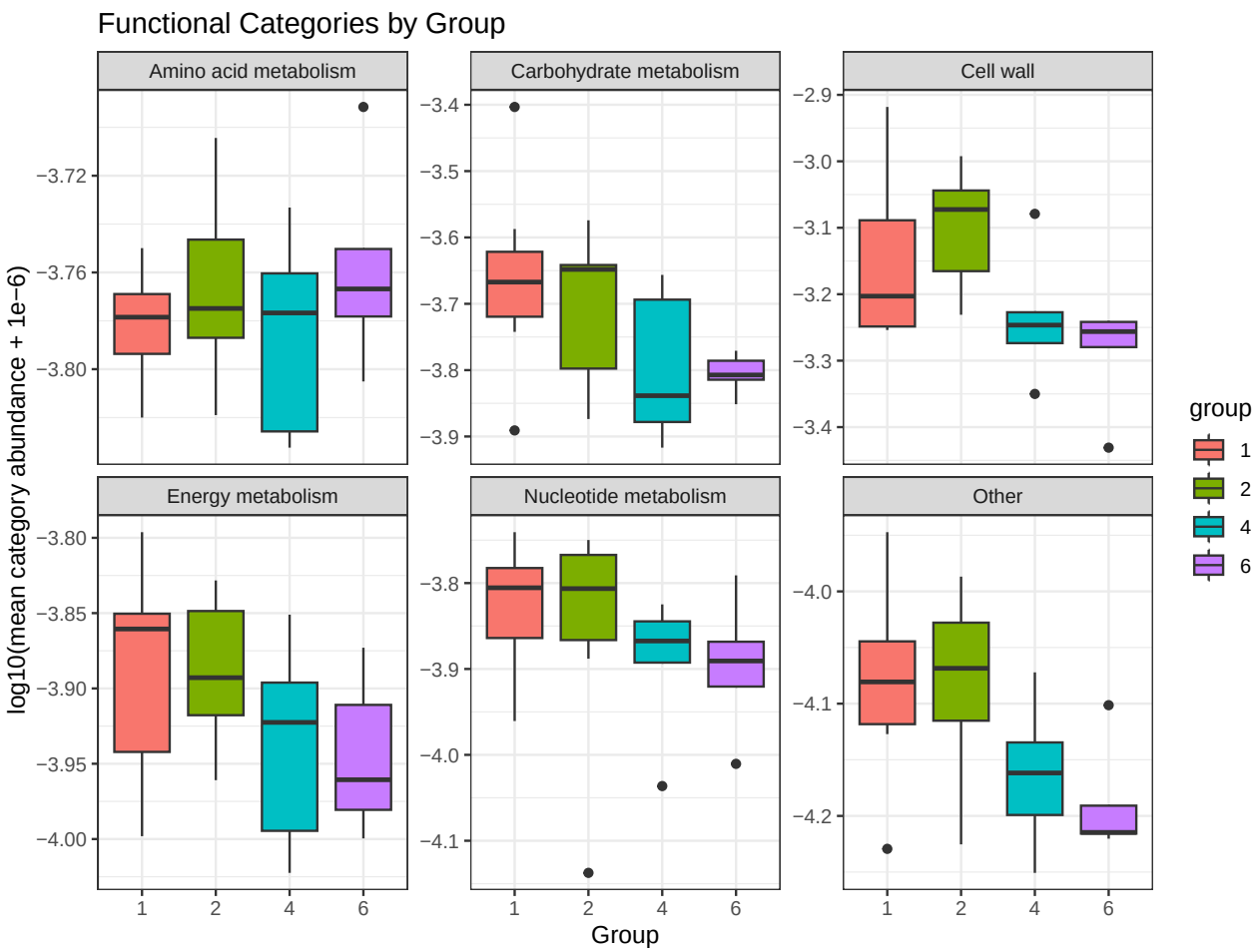


Interpretation prompt:
This aggregation is more interpretable than individual features and makes it easier to discuss pathways in terms of metabolism, cell wall processes, and nucleotide or energy functions.

Question 4: Do broad biological categories differ across groups?

```
# ============================================
# CATEGORY ABUNDANCE BY GROUP
# ============================================

cat_df_group <- as.data.frame(t(cat_mat_mean))
cat_df_group$group <- meta$group

cat_long <- cat_df_group %>%
  pivot_longer(-group, names_to = "category", values_to = "abundance")

ggplot(cat_long, aes(x = group, y = log10(abundance + 1e-6), fill = group)) +
  geom_boxplot() +
  facet_wrap(~category, scales = "free_y") +
  theme_bw() +
  labs(
    title = "Functional Categories by Group",
```

```
    x = "Group",
    y = "log10(mean category abundance + 1e-6)"
  )
```



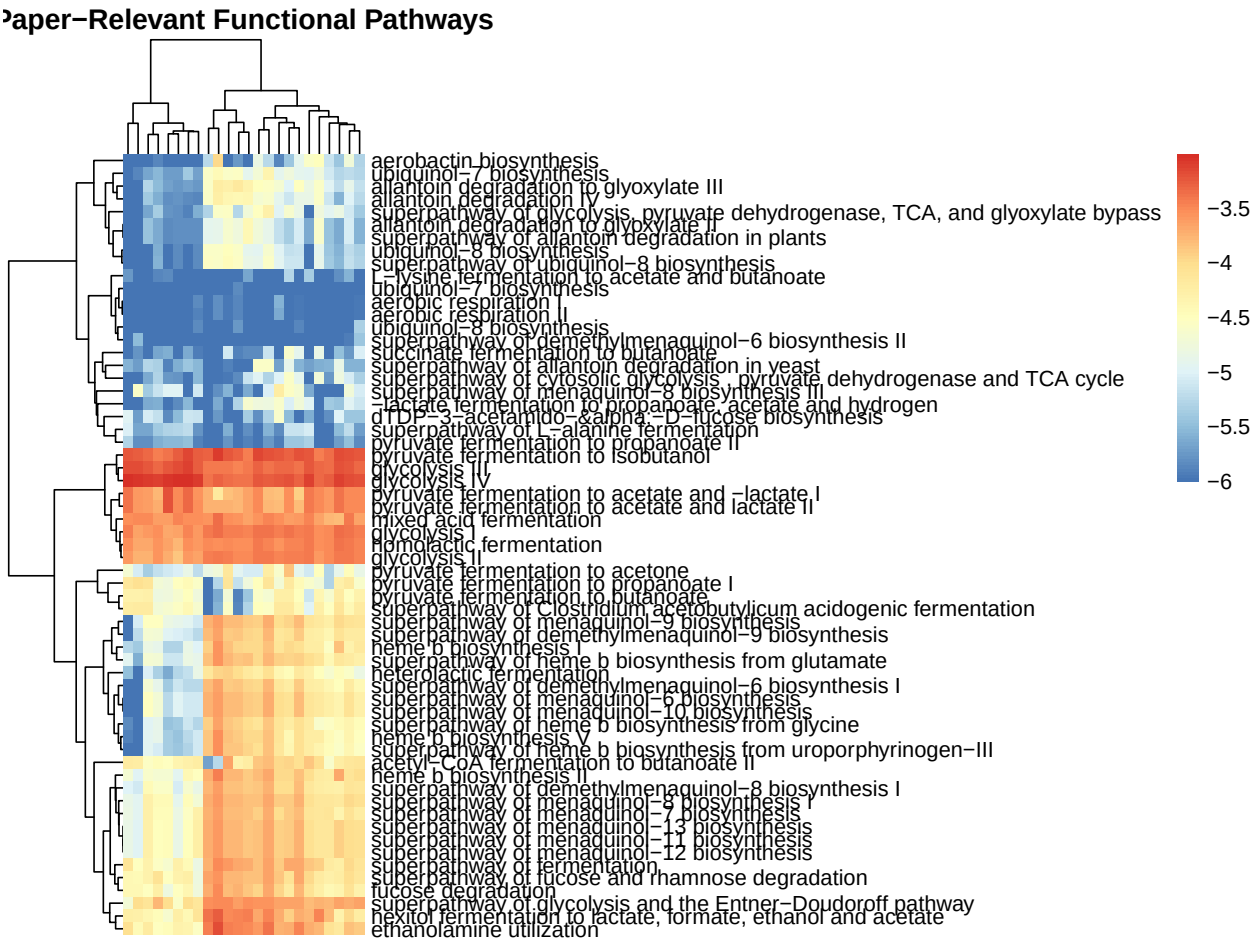
Interpretation prompt:
Do certain groups show relatively higher energy metabolism, nucleotide metabolism, or cell-wall-associated pathways? These plots summarize broad biology before moving into feature-level testing.

Question 5: Do we see paper-relevant functional themes?

The paper highlights metabolism, respiration, iron/heme handling, and nutrient utilization. We can inspect those themes directly.

```
# ==========================================
# PAPER-FOCUSED PATHWAYS
# ==========================================

clean_features <- clean_names(rownames(path_mat))

target_idx <- grepl(
  "respiration|iron|heme|glycolysis|fermentation|aerobactin|fucose|sialic|allantoin|ethanolamine|ubiqui
  clean_features,
```

```r
  ignore.case = TRUE
)

target_mat <- path_mat[target_idx, , drop = FALSE]
rownames(target_mat) <- clean_names(rownames(target_mat))
target_log <- log10(target_mat + 1e-6)

if (nrow(target_log) > 1) {
  pheatmap(
    target_log,
    show_rownames = TRUE,
    show_colnames = FALSE,
    main = "Paper-Relevant Functional Pathways"
  )
}
```



Interpretation prompt:
Are these functions consistently elevated in some groups? Do they suggest microbial communities adapted to altered nutrient availability, host-derived substrates, or stressed gut environments?

Question 6: What happens at the gene-family level?

Gene-family tables are much larger and more granular than pathways, so filtering is essential.

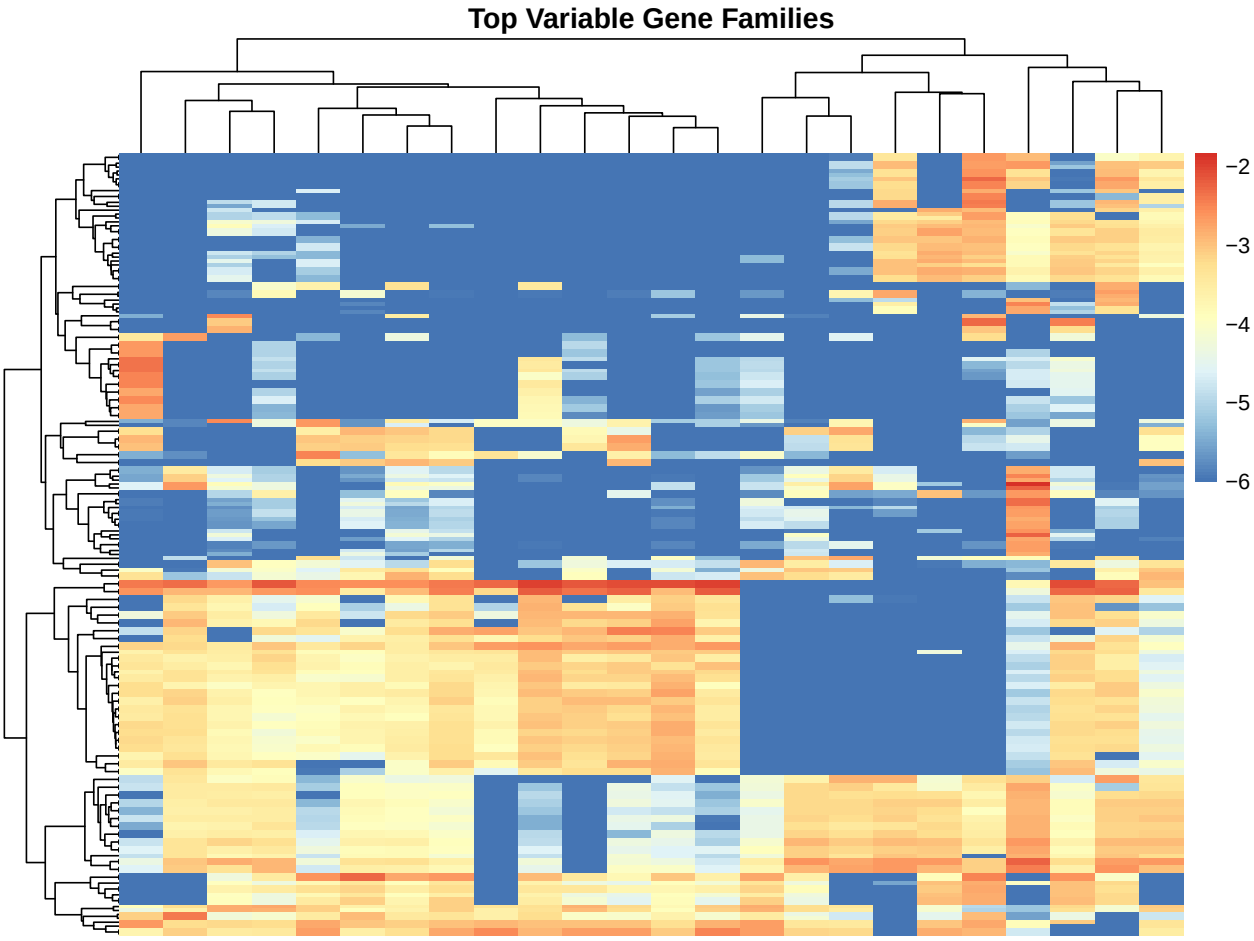
```r
# ============================================
# GENE FAMILY ANALYSIS
# ============================================

genefam_mat <- genefam %>%
  column_to_rownames("Feature") %>%
  as.matrix()

mode(genefam_mat) <- "numeric"

gene_prev <- rowMeans(genefam_mat > 0, na.rm = TRUE)
gene_mean <- rowMeans(genefam_mat, na.rm = TRUE)

genefam_filt <- genefam_mat[
  gene_prev >= 0.10 & gene_mean >= 1e-05,
  ,
  drop = FALSE
]

cat("Original gene families:", nrow(genefam_mat), "\n")
```

```
## Original gene families: 3278639
```

```r
cat("Retained after prevalence + mean-abundance filtering:", nrow(genefam_filt), "\n")
```

```
## Retained after prevalence + mean-abundance filtering: 18274
```

```r
# ============================================
# TOP VARIABLE FILTERED GENE FAMILIES
# ============================================

var_genes <- apply(genefam_filt, 1, var, na.rm = TRUE)

n_keep <- min(200, length(var_genes))
top_gene_names <- names(sort(var_genes, decreasing = TRUE))[1:n_keep]

genefam_top <- genefam_filt[top_gene_names, , drop = FALSE]
genefam_log <- log10(genefam_top + 1e-6)

pheatmap(
  genefam_log,
  show_rownames = FALSE,
  show_colnames = FALSE,
  main = "Top Variable Gene Families"
)
```

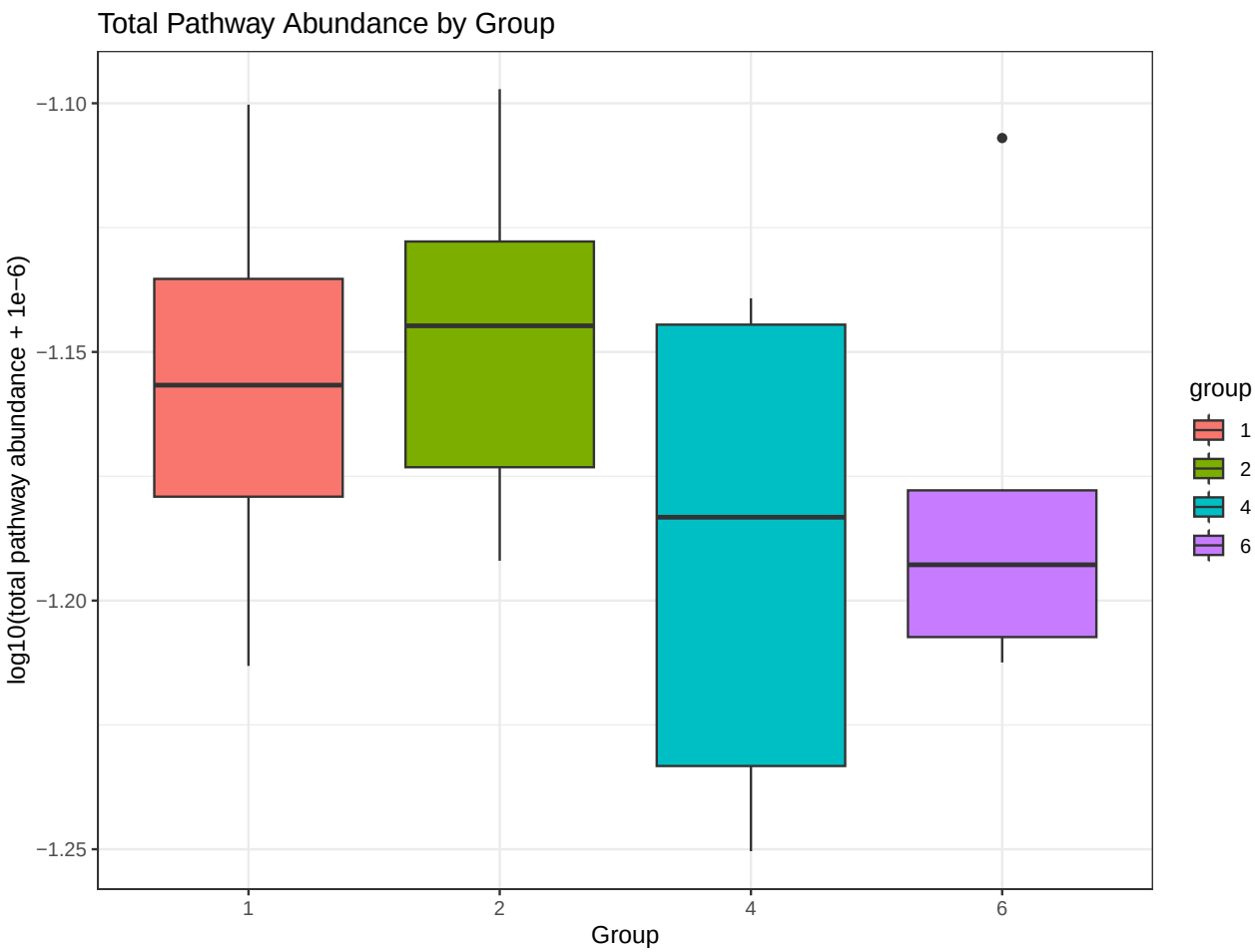


Interpretation prompt:
Gene families offer higher resolution, but they are harder to interpret biologically without annotation or targeted grouping.

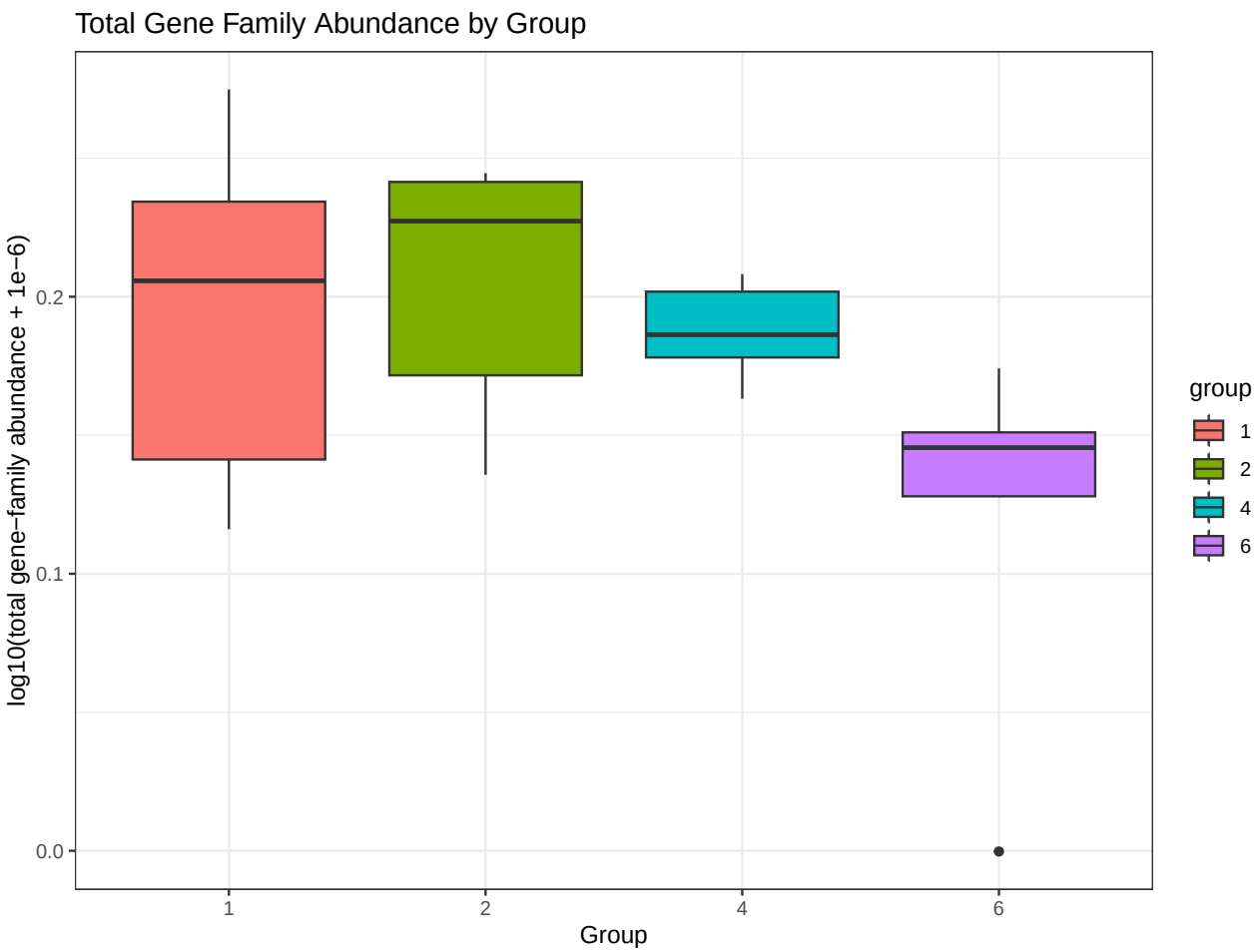
Question 7: Are total pathway or gene-family loads different across groups?

```
# ============================================
# GLOBAL FUNCTIONAL LOAD
# ============================================

meta$total_pathway_abundance <- colSums(path_mat, na.rm = TRUE)
meta$total_gene_family_abundance <- colSums(genefam_mat, na.rm = TRUE)

p1 <- ggplot(meta, aes(x = group, y = log10(total_pathway_abundance + 1e-6), fill = group)) +
  geom_boxplot() +
  theme_bw() +
  labs(
    title = "Total Pathway Abundance by Group",
    x = "Group",
    y = "log10(total pathway abundance + 1e-6)"
  )
```

```r
p2 <- ggplot(meta, aes(x = group, y = log10(total_gene_family_abundance + 1e-6), fill = group)) +
  geom_boxplot() +
  theme_bw() +
  labs(
    title = "Total Gene Family Abundance by Group",
    x = "Group",
    y = "log10(total gene-family abundance + 1e-6)"
  )

p1
```




```r
p2
```



Interpretation prompt:
These plots ask whether groups differ in the overall amount of annotated functional signal, before focusing on individual features.

Question 8: Which pathways are significantly associated with group?

```
# ============================================
# PREPARE INPUTS FOR MaAsLin2
# ============================================

path_mat_maaslin <- remove_zero_variance_features(path_mat)
genefam_top_maaslin <- remove_zero_variance_features(genefam_top)

path_maaslin <- as.data.frame(t(path_mat_maaslin))
gene_maaslin <- as.data.frame(t(genefam_top_maaslin))

stopifnot(all(rownames(path_maaslin) == meta$sample_name))
stopifnot(all(rownames(gene_maaslin) == meta$sample_name))

meta_maaslin <- meta
rownames(meta_maaslin) <- meta_maaslin$sample_name
meta_maaslin$group <- as.factor(meta_maaslin$group)
```

```r
stopifnot(!any(is.na(meta_maaslin$group)))
stopifnot(!any(duplicated(rownames(meta_maaslin))))

table(meta_maaslin$group)
```

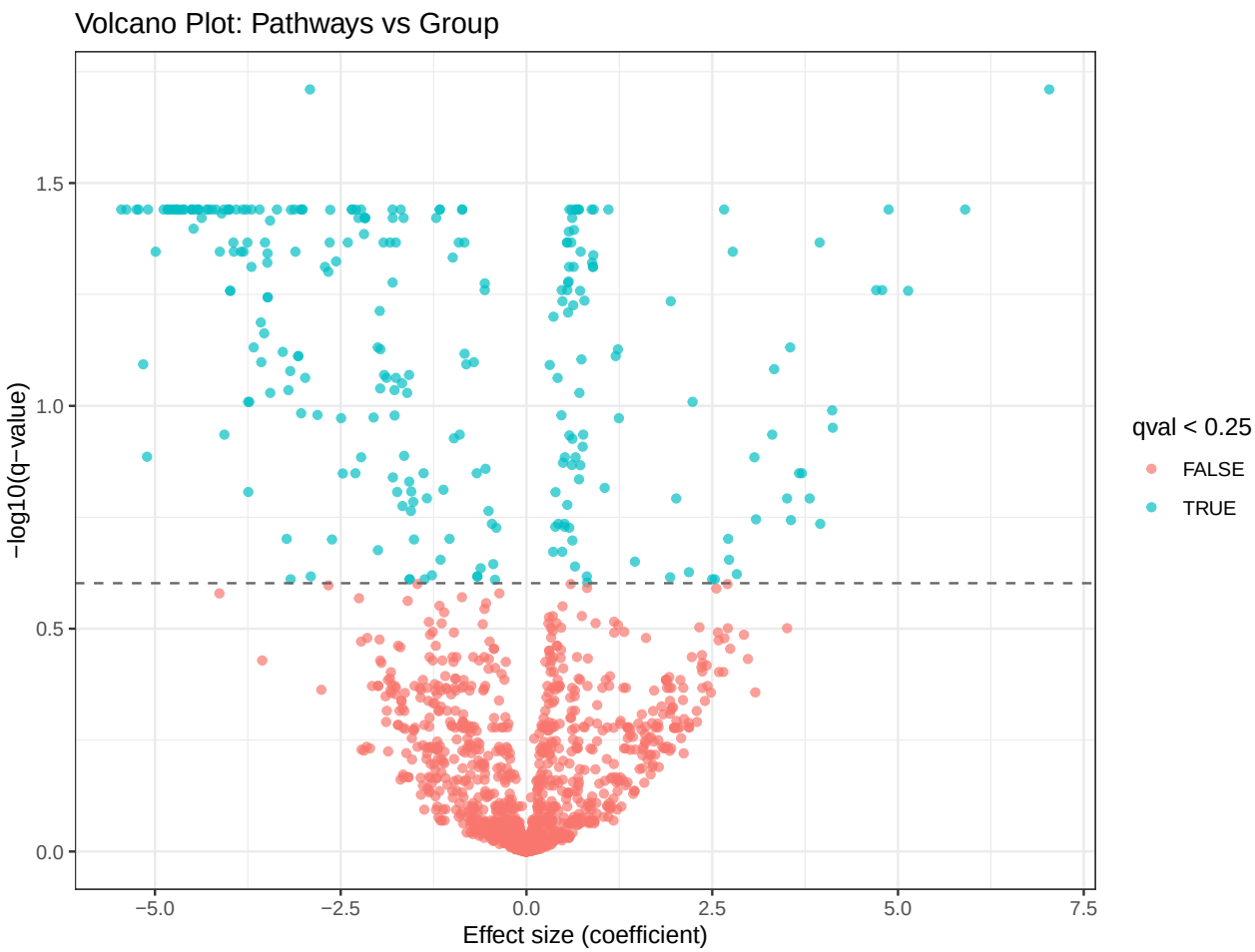
```
##
## 1 2 4 6
## 7 7 5 5
```

```r
# ============================================
# MaAsLin2: PATHWAYS
# ============================================

fit_path <- Maaslin2(
  input_data = path_maaslin,
  input_metadata = meta_maaslin,
  output = "maaslin_pathways_group",
  fixed_effects = c("group"),
  normalization = "NONE",
  transform = "LOG"
)
```

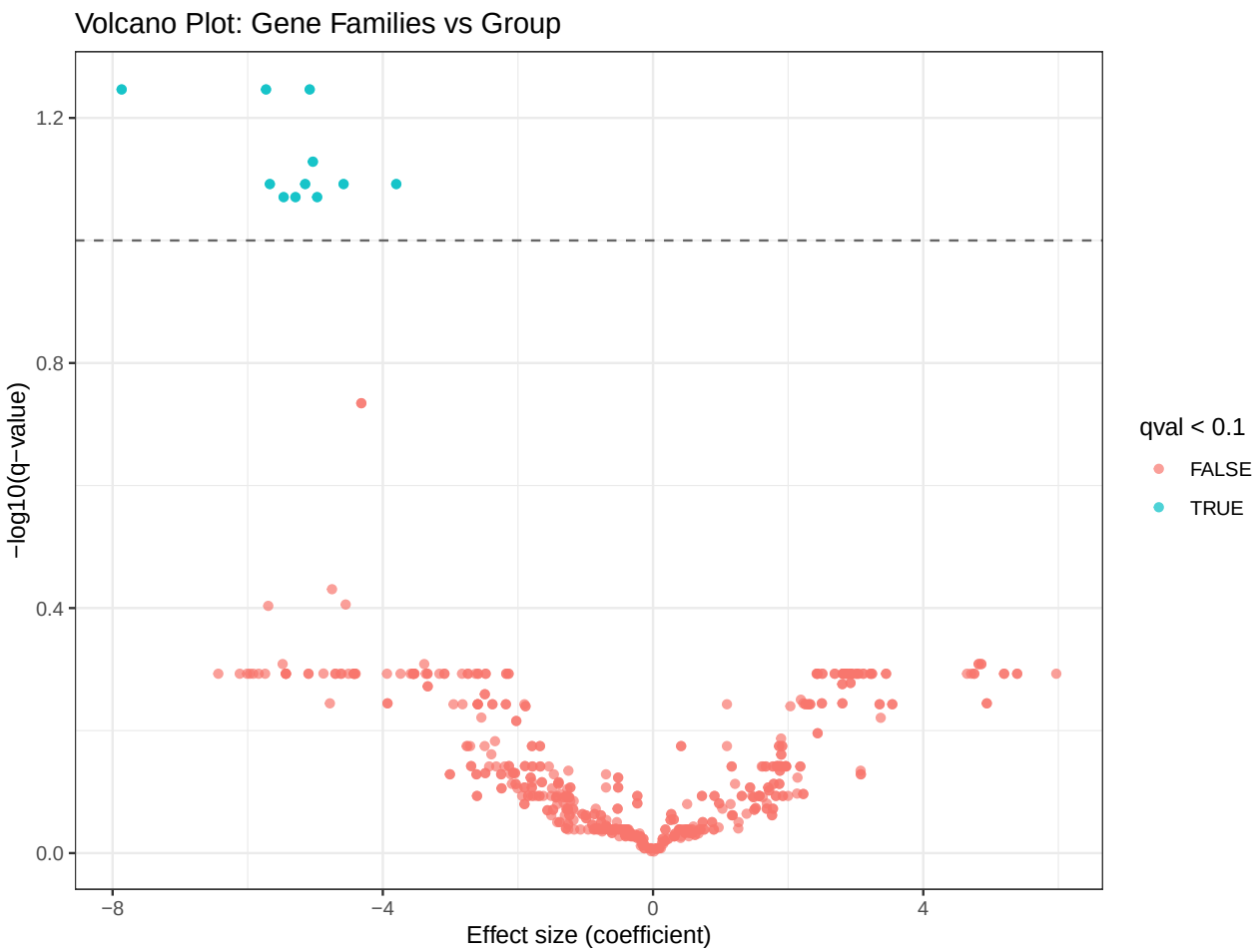
Note: These models are pre-run. Results are read from the output folders.

```r
# ============================================
# MaAsLin2: GENE FAMILIES
# ============================================

fit_gene <- Maaslin2(
  input_data = gene_maaslin,
  input_metadata = meta_maaslin,
  output = "maaslin_genefamilies_group",
  fixed_effects = c("group"),
  normalization = "NONE",
  transform = "LOG"
)
```

```r
# ============================================
# VOLCANO PLOT: PATHWAYS
# ============================================

res_path <- read.delim("maaslin_pathways_group/all_results.tsv")
res_path$clean_feature <- clean_names(res_path$feature)

ggplot(res_path, aes(x = coef, y = -log10(qval))) +
  geom_point(aes(color = qval < 0.25), alpha = 0.7) +
  geom_hline(yintercept = -log10(0.25), linetype = "dashed", color = "grey40") +
  theme_bw() +
  labs(
    title = "Volcano Plot: Pathways vs Group",
    x = "Effect size (coefficient)",
    y = "-log10(q-value)"
  )
```




```r
# ==========================================
# VOLCANO PLOT: GENE FAMILIES
# ==========================================

res_gene <- read.delim("maaslin_genefamilies_group/all_results.tsv")
res_gene$clean_feature <- clean_names(res_gene$feature)

ggplot(res_gene, aes(x = coef, y = -log10(qval))) +
  geom_point(aes(color = qval < 0.10), alpha = 0.7) +
  geom_hline(yintercept = -log10(0.10), linetype = "dashed", color = "grey40") +
  theme_bw() +
  labs(
    title = "Volcano Plot: Gene Families vs Group",
    x = "Effect size (coefficient)",
    y = "-log10(q-value)"
  )
```



Question 9: Which significant pathway results are easiest to interpret biologically?

```
# ==========================================
# SIGNIFICANT PATHWAY RESULTS
# ==========================================

res_path <- read.delim("maaslin_pathways_group/all_results.tsv", check.names = FALSE)

# Clean labels for display only
res_path <- res_path %>%
  mutate(
    clean_feature = gsub("\\.+", " ", feature) %>% trimws()
  )

# Summary first (teaching moment)
cat("Min q-value:", min(res_path$qval, na.rm = TRUE), "\n")
```

```
## Min q-value: 0.01949445
```

```r
cat("q < 0.25:", sum(res_path$qval < 0.25, na.rm = TRUE), "\n")
```

```
## q < 0.25: 267
```

```r
cat("q < 0.10:", sum(res_path$qval < 0.10, na.rm = TRUE), "\n")
```

```
## q < 0.10: 175
```

```r
cat("q < 0.05:", sum(res_path$qval < 0.05, na.rm = TRUE), "\n")
```

```
## q < 0.05: 116
```

```r
# Significant set and annotate biology
sig_path <- res_path %>%
  filter(qval < 0.25) %>%
  arrange(qval) %>%
  mutate(category = annotate_pathways(clean_feature))

# PRINT explicitly
knitr::kable(
  sig_path %>%
    select(clean_feature, category, coef, pval, qval) %>%
    slice_head(n = 20),
  caption = "Top 20 pathways associated with group"
)
```

Table 1: Top 20 pathways associated with group

clean_feature	category	coef	pval	qval
ORNDEG PWY superpathway of ornithine degradation	Amino acid metabolism	-2.9128586	0.0000270	0.0194945
PWY 622 starch biosynthesis	Amino acid metabolism	7.0361392	0.0000275	0.0194945
X1CMET2 PWY folate transformations III E coli	Other	0.5801660	0.0000903	0.0362552
ARGDEG PWY superpathway of L arginine putrescine and 4 aminobutanoate degradation	Other	-4.8270501	0.0011470	0.0362552
AST PWY L arginine degradation II AST pathway	Other	-4.6046349	0.0012957	0.0362552
CARNMET PWY L carnitine degradation I	Other	-4.7164646	0.0007401	0.0362552
COMPLETE ARO PWY superpathway of aromatic amino acid biosynthesis	Amino acid metabolism	0.7039349	0.0015377	0.0362552
DARABCATK12 PWY D arabinose degradation I	Other	-4.5037277	0.0006041	0.0362552
DENOVOPURINE2 PWY superpathway of purine nucleotides de novo biosynthesis II	Amino acid metabolism	-3.5919033	0.0009504	0.0362552
ECASYN PWY enterobacterial common antigen biosynthesis	Amino acid metabolism	-4.1776975	0.0016131	0.0362552
FAO PWY fatty acid beta oxidation I generic	Other	-4.6189535	0.0008138	0.0362552

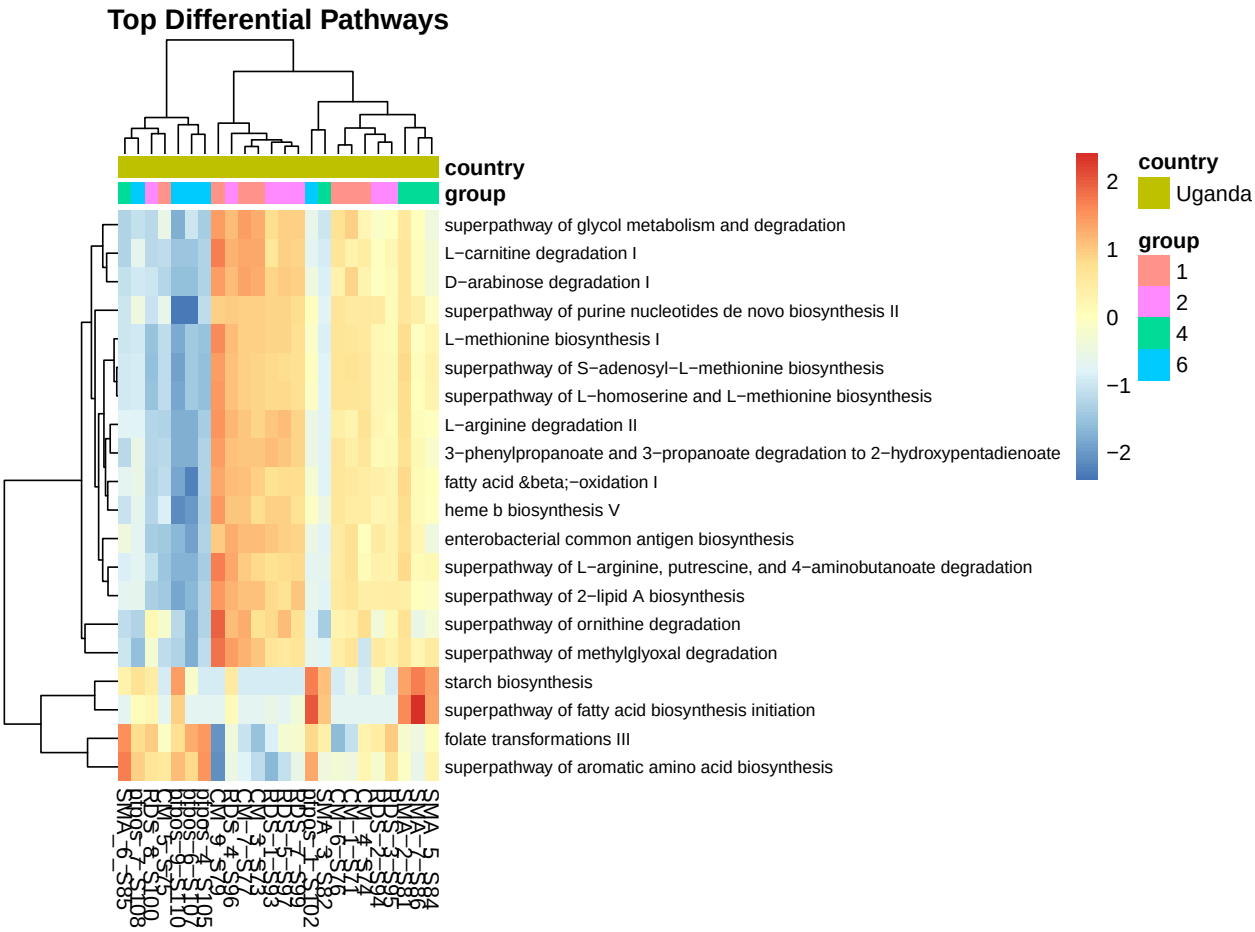
clean_feature	category	coef	pval	qval
FASYN INITIAL PWY superpathway of fatty acid biosynthesis initiation E coli	Amino acid metabolism	5.9039585	0.0006319	0.0362552
GLYCOL GLYOXDEG PWY superpathway of glycol metabolism and degradation	Other	-4.2820179	0.0001584	0.0362552
HCAMHPDEG PWY 3 phenylpropanoate and 3 3 hydroxyphenyl propanoate degradation to 2 hydroxypentadienoate	Other	-4.5010399	0.0014518	0.0362552
HEME BIOSYNTHESIS II 1 heme b biosynthesis V aerobic	Amino acid metabolism	-5.2472796	0.0005017	0.0362552
HOMOSER METSYN PWY L methionine biosynthesis I	Amino acid metabolism	-3.3573764	0.0012924	0.0362552
KDO NAGLIPASYN PWY superpathway of Kdo 2 lipid A biosynthesis	Amino acid metabolism	-4.4074322	0.0010902	0.0362552
MET SAM PWY superpathway of S adenosyl L methionine biosynthesis	Amino acid metabolism	-3.0124419	0.0013502	0.0362552
METHGLYUT PWY superpathway of methylglyoxal degradation	Other	-2.3449613	0.0009486	0.0362552
METSYN PWY superpathway of L homoserine and L methionine biosynthesis	Amino acid metabolism	-3.1231568	0.0013168	0.0362552

```r
# ============================================
# HEATMAP OF TOP DIFFERENTIAL PATHWAYS
# ============================================

# Map MaAsLin2 feature names back to original HUMAnN names
feature_map <- data.frame(
  original = rownames(path_mat),
  maaslin = make.names(rownames(path_mat)),
  stringsAsFactors = FALSE
)

top_maaslin <- sig_path %>%
  slice_head(n = 20) %>%
  pull(feature)

top_original <- feature_map$original[
  match(top_maaslin, feature_map$maaslin)
]

top_original <- na.omit(top_original)

cat("Mapped features:", length(top_original), "\n")
```

```
## Mapped features: 20
```

```r
if (length(top_original) >= 2) {

  abund_top <- path_mat[top_original, , drop = FALSE]
  rownames(abund_top) <- clean_names(rownames(abund_top))

  anno <- meta %>%
```

```
  select(sample_name, group, country) %>%
  as.data.frame()

rownames(anno) <- anno$sample_name
anno$sample_name <- NULL

  pheatmap(
    log10(abund_top + 1e-6),
    annotation_col = anno,
    scale = "row",
    fontsize_row = 8,
    main = "Top Differential Pathways"
  )

} else {
  cat("Not enough features for heatmap\n")
}
```



Interpretation prompt:
Do the significant features point to broad metabolic shifts, or to specific processes such as respiration, iron handling, or nutrient scavenging?

Question 10: Can we collapse gene families into targeted biological modules?

This section turns gene families into interpretable functional modules inspired by the paper.

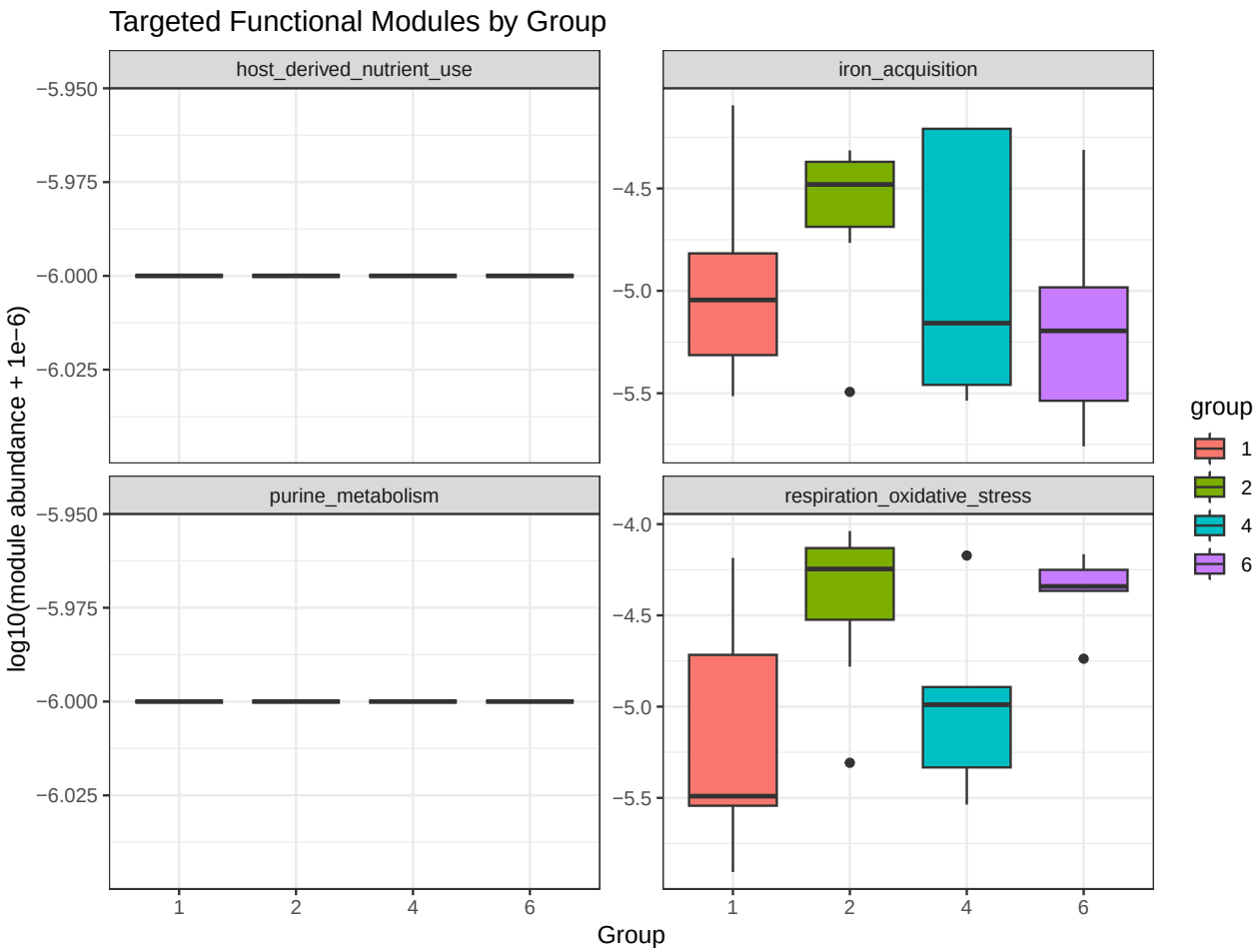
```
# ============================================
# TARGETED GENE-FAMILY MODULES
# ============================================

gene_ids <- rownames(genefam_mat)

bucket <- rep("other", length(gene_ids))
bucket[grepl("aerobactin|siderophore|fhu|heme", gene_ids, ignore.case = TRUE)] <- "iron_acquisition"
bucket[grepl("formate dehydrogenase|fdn|ubiquinone|menaquinone|superoxide dismutase", gene_ids, ignore.c
bucket[grepl("purine|adenosine deaminase|allantoin", gene_ids, ignore.case = TRUE)] <- "purine_metaboli
bucket[grepl("ethanolamine|fucose|sialic", gene_ids, ignore.case = TRUE)] <- "host_derived_nutrient_use

table(bucket)
```

```
## bucket
##             iron_acquisition                        other
##                           75                      3278448
## respiration_oxidative_stress
##                          116
```

```
iron_total <- colSums(genefam_mat[bucket == "iron_acquisition", , drop = FALSE], na.rm = TRUE)
resp_total <- colSums(genefam_mat[bucket == "respiration_oxidative_stress", , drop = FALSE], na.rm = TRU
purine_total <- colSums(genefam_mat[bucket == "purine_metabolism", , drop = FALSE], na.rm = TRUE)
hostnut_total <- colSums(genefam_mat[bucket == "host_derived_nutrient_use", , drop = FALSE], na.rm = TRU

target_df <- data.frame(
  sample_name = colnames(genefam_mat),
  iron_acquisition = as.numeric(iron_total),
  respiration_oxidative_stress = as.numeric(resp_total),
  purine_metabolism = as.numeric(purine_total),
  host_derived_nutrient_use = as.numeric(hostnut_total)
) %>%
  left_join(meta, by = "sample_name")
```

```
# ============================================
# TARGETED MODULES BY GROUP
# ============================================

target_long <- target_df %>%
  pivot_longer(
    cols = c(iron_acquisition, respiration_oxidative_stress, purine_metabolism, host_derived_nutrient_us
    names_to = "module",
    values_to = "abundance"
  )

ggplot(target_long, aes(x = group, y = log10(abundance + 1e-6), fill = group)) +
  geom_boxplot() +
  facet_wrap(~module, scales = "free_y") +
  theme_bw() +
```

```r
  labs(
    title = "Targeted Functional Modules by Group",
    x = "Group",
    y = "log10(module abundance + 1e-6)"
  )
```



Interpretation prompt:
Do these targeted functional modules recapitulate the biology emphasized in the paper?

Reproducibility

To run this module:

1. Clone the repository
2. Place input files in the `data/` directory
3. Open the `.Rmd` file in RStudio
4. Click **Knit → PDF**

Note: MaAsLin2 outputs are precomputed.